

Experiment 1: Doctor Shift Assignment using Backtracking

Aim

To assign three doctors to three hospital shifts using the Backtracking Search algorithm while satisfying all given constraints.

Problem Statement

Assign three doctors (D1, D2, D3) to Morning, Afternoon, and Night shifts such that:

D1 cannot work the Night shift.

D2 must work before D3.

Only one doctor is assigned to each shift.

D3 cannot work the Morning shift.

Every doctor is assigned exactly one shift.

Solution

Assign D1 → Morning

Assign D2 → Afternoon

Assign D3 → Night

Constraint Checks

D1 is not in Night ✓

D2 works before D3 ✓

D3 is not in Morning ✓

One doctor per shift ✓

All doctors assigned ✓

Result

A valid schedule is obtained:

Morning → D1

Afternoon → D2

Night → D3

Experiment 2: Robot Path Finding using BFS

Aim

To find the shortest path from the Start node to the Goal node using the Breadth-First Search (BFS) algorithm.

Problem Statement

A robot moves in a 5×5 grid from Start (S) to Goal (G). It can move Up, Down, Left, or Right. Obstacles cannot be crossed, and each move has a cost of 1.

Solution

Start from S.

Expand neighboring nodes level by level.

Ignore obstacle cells.

Continue until Goal (G) is reached.

Optimal Path

S → Right → Right → Down → Down → Right → Right → Down → G

Cost

Total Moves = 8

Total Cost = 8

Result

The robot reaches the goal through the shortest valid path.

Experiment 3: Autonomous Rescue Robot

Aim

To design a rescue robot that safely reaches a survivor using an appropriate AI search strategy.

Problem Statement

A rescue robot operates in a partially observable building. It must reach a survivor while avoiding obstacles and minimizing movement cost.

Solution

Observe adjacent cells.

Avoid blocked paths.

Calculate movement costs.

Choose the lowest-cost path using Uniform Cost Search or an Online Search Agent.

Update knowledge dynamically while moving.

AI Concepts

Intelligent Agent

Rational Decision Making

Partially Observable Environment

Online Search

Uniform Cost Search

Result

The robot successfully reaches the survivor by selecting the safest and minimum-cost path.